

PUNIT MISHRA

Senior Machine Learning Expert & AI Tech Lead — Agents, Inference & Multi-Tenant Platforms at Enterprise Scale

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PROFILE

Applied AI engineer and technical leader with 14+ years taking machine learning from prototype to dependable production at enterprise scale. Not only an architect — I **implement, deploy, and own systems end to end**, from agent runtimes and LLM inference pipelines through **GPU deployment on custom H100 clusters**. As Senior ML Expert and AI Tech Lead in SAP CX Data & AI, I lead a team of thirteen *and personally build* the agent orchestration, memory, multi-model serving, and multi-tenant infrastructure beneath our LLM products. Deep command of Rust, Kubernetes, distributed inference, vector search, and AI security.

EXPERIENCE

SAP Sep 2018 – Present
Senior Machine Learning Expert & AI Tech Lead — CX Data & AI *San Ramon, CA*

- **Built and own a domain-agnostic conversational agent platform** — a memory-managed runtime with multi-tenant LangGraph orchestration reusable across business domains, with performance-critical paths written in **Rust**.
- Authored **natural-language-to-SQL over S/4HANA**, turning business questions into validated queries — now shipping within SAP's **2026 public and private cloud releases**.
- **Introduced Ray** and built **LLM serving on custom bare-metal H100 GPU clusters**; multi-model serving (self-hosted plus Claude, Qwen, Granite-class) on Kubernetes with automated evaluation harnesses.
- Delivered visual and textual LLM agents across SAP's **CX & Commerce AI Toolkit** (PyTorch on Kubernetes); visual-attribute prediction on OpenAI / CLIP embeddings backed by **vector search and RAG**.
- Designed **multi-tenant ML** on Kubeflow & Knative (Azure AKS) across SAP CX cloud (C4C, CPQ, CDC), with feature stores (Feathr, Feast); **lead a team of thirteen**, own AI ethics & compliance, and set the group's SDLC and engineering standards. **SAP Security Expert**.
- Re-architected **vision** — attribute extraction, object detection, visual search — into first-class **gRPC agent services** exposed through an agent gateway and MCP-based service hub.
- Designed multi-tenant identity and provisioning unifying **IAS and XSUAA**; established the CI/CD backbone with **GitOps promotion** (Argo / Piper) on Kubernetes / Kyma (Gardener).
- Serve as **forward-deployed engineer** on the platform's first production enterprise customer, deploying always-on agent services at very large scale across multiple customers.

CALLIDUSCLOUD Feb 2014 – Sep 2018
Senior Software Engineer — Thunderbridge AI / SAP Sales Cloud *SF Bay Area (acquired by SAP, \$2.4B)*

- Designed and built the **Thunderbridge AI** backend in Python on **Dataiku**, with multi-tenancy running every customer workload for SAP Sales Cloud optimization.
- Ported the full Dataiku ML stack to **GKE on-prem (Google Anthos)** with Istio, and built the CI/CD backbone — Jenkins on Kubernetes, Kaniko image builds, Artifactory — plus automated merges (Palantir Bulldozer / Policy-Bot).
- Engineered multi-partition data resiliency for ML across **Oracle, Vertica, SAP HANA, and HDFS** (Spark); shipped inference services for CPQ on Dataiku's API node.
- Integrated SAP security scans (**Coverity, Black Duck, Fortify**) into Jenkins per commit. As Senior Automation Engineer (earlier), built TDD/Scala ETL validation and resilient Selenium/Java and Node.js UI test frameworks over a self-maintaining Selenium Grid.

IBM Oct 2012 – Dec 2013

- Built a reporting and **data-validation automation framework** for JPMorgan Chase.
- Automated VM creation/configuration via a WCF SOAP service (IIS); parallelized test execution and scheduled runs across cloud machines with varied OS/browser configs.
- Developed reusable **Selenium / Java** test APIs (JUnit, TestNG) adopted across the firm.

UC BERKELEY LAW IST

Sep – Dec 2011

*Web Developer**Berkeley, CA*

- Built an online teaching-evaluation system for the Law School on a PHP / Oracle 10g backend, integrated with **CalNet** (Berkeley authentication) for end-of-course student evaluations.

LAWPIVOT INC. (GOOGLE STARTUP LABS)

May – Aug 2011

*Software Engineering Intern**Mountain View, CA*

- Built user-to-user interaction features on the Django backend and an internal analytics tool visualizing user data (logins, time on site) used by the sales team.

LAWPIVOT INC.

Oct 2009 – Aug 2010

*Software Engineering Intern**Fremont, CA*

- Automated testing with Selenium IDE and tracked defects; worked with the founding team on wireframes and AWS server configuration.

SKILLS

AI & AGENTS	Agent orchestration (LangGraph, LangChain), MCP (agent gateway / tool hub), LLM serving & evaluation, NL-to-SQL, RAG, retriever–reranker, PyTorch, OpenAI / CLIP, MLOps (Kubeflow, Feathr, Feast)
GPU & SERVING	Ray (distributed), custom bare-metal H100 clusters, self-hosted LLM / embedding serving, Knative / KServe
INFRA & CLOUD	Kubernetes, Kyma / Gardener, Istio, Argo / GitOps, Jenkins, Docker, CI/CD; vector DBs (Qdrant, FAISS), Redis; Azure (AKS), GCP (GKE / Anthos)
PLATFORM & DATA	gRPC / protobuf, CAP / XSUAA (OIDC / PKCE); PostgreSQL, SAP HANA, Oracle, Vertica, HDFS / Spark, Kafka
LANGUAGES	Python (FastAPI, Django), Rust, Java, Scala, TypeScript / Next.js
SECURITY	SAP Security Expert; internal-auth / identity design, threat modeling, SAST / SCA (Coverity, Black Duck, Fortify)

EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY — B.S. Electrical Engineering & Computer Science

2012